

Practical Session 2

Task 1

Develop a C program that uses the system calls `open()`, `read()`, `write()`, and `close()` to copy the contents of one file to another. Explain how control transitions between user space and kernel space during execution.

```
j_rakshitha@RAKSHITHA:~/ossp_project$ nano src/file_copy.c
j_rakshitha@RAKSHITHA:~/ossp_project$ gcc -Wall -Wextra -g src/file_copy.c -o build/file_copy
j_rakshitha@RAKSHITHA:~/ossp_project$ echo "This is a sample file for the OSSP file copy task." > sample.txt
j_rakshitha@RAKSHITHA:~/ossp_project$ cat sample.txt
This is a sample file for the OSSP file copy task.
j_rakshitha@RAKSHITHA:~/ossp_project$ ./build/file_copy sample.txt copied.txt
File copied successfully.
Source      : sample.txt
Destination : copied.txt
j_rakshitha@RAKSHITHA:~/ossp_project$ cat copied.txt
This is a sample file for the OSSP file copy task.
j_rakshitha@RAKSHITHA:~/ossp_project$ cmp sample.txt copied.txt
j_rakshitha@RAKSHITHA:~/ossp_project$ |
```

```
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <stdlib.h>

#define BUFFER_SIZE 1024

int main(int argc, char *argv[]) {
    if (argc != 3) {
        printf("Usage: %s <source_file> <destination_file>\n", argv[0]);
        return 1;
    }

    int source_fd = open(argv[1], O_RDONLY);

    if (source_fd < 0) {
        perror("Error opening source file");
        return 1;
    }

    int destination_fd = open(
        argv[2],
        O_WRONLY | O_CREAT | O_TRUNC,
        0644
    );

    if (destination_fd < 0) {
        perror("Error opening destination file");
        close(source_fd);
        return 1;
    }

    char buffer[BUFFER_SIZE];
    ssize_t bytes_read;

    while ((bytes_read = read(source_fd, buffer, BUFFER_SIZE)) > 0) {
        ssize_t bytes_written = write(
            destination_fd,
            buffer
```

```
);

if (destination_fd < 0) {
    perror("Error opening destination file");
    close(source_fd);
    return 1;
}

char buffer[BUFFER_SIZE];
ssize_t bytes_read;

while ((bytes_read = read(source_fd, buffer, BUFFER_SIZE)) > 0) {

    ssize_t bytes_written = write(
        destination_fd,
        buffer,
        bytes_read
    );

    if (bytes_written != bytes_read) {
        perror("Error writing to destination file");
        close(source_fd);
        close(destination_fd);
        return 1;
    }
}

if (bytes_read < 0) {
    perror("Error reading source file");
}

close(source_fd);
close(destination_fd);

printf("File copied successfully.\n");
printf("Source      : %s\n", argv[1]);
printf("Destination : %s\n", argv[2]);

return 0;
}
```

Task 2

Use the strace utility to trace all system calls generated by the command cat sample.txt. Analyze the sequence of system calls and identify the kernel services involved.

```
j_rakshitha@RAKSHITHA:~/ossp_project$ strace -V
Command 'strace' not found, but can be installed with:
sudo apt install strace
j_rakshitha@RAKSHITHA:~/ossp_project$ sudo apt install strace
Installing:
  strace

Installing dependencies:
  libunwind8

Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 40
  Download size: 737 kB
  Space needed: 2660 kB / 1024 GB available

Continue? [Y/n] Y
Get:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 libunwind8 amd64 1.8.3-0ubuntu1 [59.6 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute/main amd64 strace amd64 6.19+ds-0ubuntu5 [678 kB]
Fetched 737 kB in 1s (514 kB/s)
Selecting previously unselected package libunwind8:amd64.
(Reading database ... 43151 files and directories currently installed.)
Preparing to unpack .../libunwind8_1.8.3-0ubuntu1_amd64.deb ...
Unpacking libunwind8:amd64 (1.8.3-0ubuntu1) ...
Selecting previously unselected package strace.
Preparing to unpack .../strace_6.19+ds-0ubuntu5_amd64.deb ...
Unpacking strace (6.19+ds-0ubuntu5) ...
Setting up libunwind8:amd64 (1.8.3-0ubuntu1) ...
Setting up strace (6.19+ds-0ubuntu5) ...
Processing triggers for man-db (2.13.1-1build1) ...
Processing triggers for libc-bin (2.43-2ubuntu2.3) ...
j_rakshitha@RAKSHITHA:~/ossp_project$ strace -V
strace -- version 6.19
Copyright (c) 1991-2026 The strace developers <https://strace.io>.
This is free software; see the source for copying conditions.  There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

Optional features enabled: stack-trace=libunwind stack-demangle m32-mpers mx32-mper
s
```

```

j_rakshitha@RAKSHITHA:~/ossp_project$ strace cat sample.txt
execve("/usr/bin/cat", ["cat", "sample.txt"], 0x7ffdc60887a8 /* 27 vars */) = 0
brk(NULL)                               = 0x58b52e244000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x738b371af000
access("/etc/ld.so.preload", R_OK)      = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=21555, ...}) = 0
mmap(NULL, 21555, PROT_READ, MAP_PRIVATE, 3, 0) = 0x738b371a9000
close(3)                                 = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=199120, ...}) = 0
mmap(NULL, 206288, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x738b37176000
mmap(0x738b3717d000, 135168, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x7000) = 0x738b3717d000
mmap(0x738b3719e000, 28672, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x738b3719e000
mmap(0x738b371a5000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2f000) = 0x738b371a5000
mmap(0x738b371a7000, 5584, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x738b371a7000
close(3)                                 = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libgcc_s.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=187120, ...}) = 0
mmap(NULL, 185256, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x738b37148000
mmap(0x738b3714c000, 147456, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x4000) = 0x738b3714c000
mmap(0x738b37170000, 16384, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x738b37170000
mmap(0x738b37174000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2c000) = 0x738b37174000
close(3)                                 = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libm.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=1198376, ...}) = 0
mmap(NULL, 1200152, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x738b37022000

```

```
TE, 3, 0x12000) = 0x738b37034000
mmap(0x738b370d4000, 466944, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0xb2000) = 0x738b370d4000
mmap(0x738b37146000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x123000) = 0x738b37146000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\3\0>\0\1\0\0\0 \250\2\0\0\0\0\0"... , 832) = 832
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@_\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"... , 840, 64) = 840
fstat(3, {st_mode=S_IFREG|0755, st_size=2186512, ...}) = 0
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@_\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"... , 840, 64) = 840
mmap(NULL, 2231696, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x738b36e00000
mmap(0x738b36e28000, 1671168, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x738b36e28000
mmap(0x738b36fc0000, 319488, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0xc0000) = 0x738b36fc0000
mmap(0x738b3700e000, 24576, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x20d000) = 0x738b3700e000
mmap(0x738b37014000, 52624, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x738b37014000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libpcr2-8.so.0", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0"... , 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=699056, ...}) = 0
mmap(NULL, 701112, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x738b36d54000
mmap(0x738b36d57000, 507904, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x3000) = 0x738b36d57000
mmap(0x738b36dd3000, 176128, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x7f000) = 0x738b36dd3000
mmap(0x738b36dfe000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0xa9000) = 0x738b36dfe000
close(3) = 0
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x738b36d52000
mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x738b36d4f000
arch_prctl(ARCH_SET_FS, 0x738b36d4fb00) = 0
set_tid_address(0x738b36d50128) = 4171
set_robust_list(0x738b36d4fde0, 24) = 0
```

```

mprotect(0x738b37146000, 4096, PROT_READ) = 0
mprotect(0x738b37174000, 4096, PROT_READ) = 0
mprotect(0x738b371a5000, 4096, PROT_READ) = 0
mprotect(0x58b4ef51c000, 1462272, PROT_READ) = 0
mprotect(0x738b371f4000, 8192, PROT_READ) = 0
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INFINITY}) =
0
getrandom("\xcd\xe3\xb2\x79\xa0\x1b\x8b\x6a", 8, GRND_NONBLOCK) = 8
munmap(0x738b371a9000, 21555) = 0
statfs("/sys/fs/selinux", {f_type=SYSFS_MAGIC, f_bsize=4096, f_blocks=0, f_bfree=0,
f_bavail=0, f_files=0, f_ffree=0, f_fsid={val=[0xe4dd8142, 0xec3f5a0e]}, f_namelen
=255, f_frsize=4096, f_flags=ST_VALID|ST_NOSUID|ST_NODEV|ST_NOEXEC|ST_NOATIME}) = 0
statfs("/selinux", 0x7ffc84c46ca0) = -1 ENOENT (No such file or directory)
brk(NULL) = 0x58b52e244000
brk(0x58b52e265000) = 0x58b52e265000
openat(AT_FDCWD, "/proc/filesystems", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0444, st_size=0, ...}) = 0
read(3, "nodev\tsysfs\nnodev\ttmpfs\nnodev\tbd"... , 1024) = 442
lseek(3, 0, SEEK_CUR) = 442
lseek(3, 416, SEEK_SET) = 416
close(3) = 0
openat(AT_FDCWD, "/proc/mounts", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0444, st_size=0, ...}) = 0
read(3, "none /usr/lib/modules/6.18.33.2-"... , 1024) = 1024
read(3, "e 0 0\ndevpts /dev/pts devpts rw,"... , 1024) = 1024
read(3, ",access=client,msize=65536,trans"... , 1024) = 1024
read(3, "700,noswap 0 0\ntmpfs /run/user/1"... , 1024) = 250
read(3, "", 1024) = 0
close(3) = 0
access("/etc/selinux/config", F_OK) = -1 ENOENT (No such file or directory)
fcntl(0, F_GETFD) = 0
fcntl(1, F_GETFD) = 0
fcntl(2, F_GETFD) = 0
rt_sigaction(SIGPIPE, NULL, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 8) = 0
poll([{fd=0, events=0}, {fd=1, events=0}, {fd=2, events=0}], 3, 0) = 0 (Timeout)
rt_sigaction(SIGPIPE, {sa_handler=SIG_IGN, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_
RESTART, sa_restorer=0x738b36e45cb0}, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0},
8) = 0
openat(AT_FDCWD, "/proc/self/maps", O_RDONLY|O_CLOEXEC) = 3
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INFINITY}) =
0
fstat(3, {st_mode=S_IFREG|0444, st_size=0, ...}) = 0
read(3, "58b4eebb3000-58b4eeca5000 r--p 0"... , 1024) = 1024
read(3, "r/lib/x86_64-linux-gnu/libpcr2-"... , 1024) = 1024

```

```

openat(AT_FDCWD, "/usr/share/coreutils/locales/uucore/en-US.ftl", O_RDONLY|O_CLOEXEC) = 3
statx(3, "", AT_STATX_SYNC_AS_STAT|AT_EMPTY_PATH, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFREG|0644, stx_size=4080, ...}) = 0
read(3, "# Common strings shared across a...", 4080) = 4080
read(3, "", 32) = 0
close(3) = 0
statx(AT_FDCWD, "/usr/share/coreutils/locales/cat", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFDIR|0755, stx_size=4096, ...}) = 0
openat(AT_FDCWD, "/usr/share/coreutils/locales/cat/en-US.ftl", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
rt_sigaction(SIGPIPE, {sa_handler=SIG_DFL, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x738b36e45cb0}, {sa_handler=SIG_IGN, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x738b36e45cb0}, 8) = 0
rt_sigaction(SIGSEGV, {sa_handler=SIG_DFL, sa_mask=~[RTMIN RT_1], sa_flags=SA_RESTORER, sa_restorer=0x738b36e45cb0}, {sa_handler=0x58b4eed80c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x738b36e45cb0}, 8) = 0
rt_sigaction(SIGBUS, {sa_handler=SIG_DFL, sa_mask=~[RTMIN RT_1], sa_flags=SA_RESTORER, sa_restorer=0x738b36e45cb0}, {sa_handler=0x58b4eed80c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x738b36e45cb0}, 8) = 0
ioctl(1, TCGETS2, {c_iflag=ICRNL|IXON, c_oflag=NL0|CR0|TAB0|BS0|VT0|FF0|OPOST|ONLCR, c_cflag=B38400|CS8|CREAD, c_lflag=ISIG|ICANON|ECHO|ECHOE|ECHOK|IEXTEN|ECHOCTL|ECHO, ...}) = 0
brk(0x58b52e286000) = 0x58b52e286000
statx(AT_FDCWD, "sample.txt", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFREG|0644, stx_size=51, ...}) = 0
openat(AT_FDCWD, "sample.txt", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=51, ...}) = 0
fstat(1, {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0x2), ...}) = 0
splice(3, NULL, 1, NULL, 1048576, 0) = -1 EINVAL (Invalid argument)
pipe([4, 5]) = 0
fcntl(4, F_SETPIPE_SZ, 1048576) = 1048576
splice(3, NULL, 5, NULL, 1048576, 0) = 51
splice(4, NULL, 1, NULL, 51, 0This is a sample file for the OSSP file copy task.
) = 51
splice(3, NULL, 5, NULL, 1048576, 0) = 0
close(5) = 0
close(4) = 0
close(3) = 0
sigaltstack({ss_sp=NULL, ss_flags=SS_DISABLE, ss_size=8192}, NULL) = 0
munmap(0x738b371ac000, 12288) = 0
exit_group(0) = ?
+++ exited with 0 +++
j_rakshitha@RAKSHITHA:~/ossp_project$

```